

Theorem (Cauchy–Schwarz). For real numbers $a_1, \dots, a_n$ and $b_1, \dots, b_n$,

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right).$$

Equality holds iff there exists $\lambda \in \mathbb{R}$ such that $a_i = \lambda b_i$ for every i (the statement is vacuous when one of the vectors is the zero vector).

Proof. **1. Interpretation as inner product.** Define the vectors

$$\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{R}^n, \quad \mathbf{b} = (b_1, \dots, b_n) \in \mathbb{R}^n.$$

The standard Euclidean inner product and norm are

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{i=1}^n a_i b_i, \quad \|\mathbf{a}\| = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle} = \sqrt{\sum_{i=1}^n a_i^2}, \quad \|\mathbf{b}\| = \sqrt{\sum_{i=1}^n b_i^2}.$$

2. A non-negative quadratic. For $t \in \mathbb{R}$ consider $\mathbf{a} + t\mathbf{b}$. Its squared norm is

$$f(t) = \|\mathbf{a} + t\mathbf{b}\|^2 = \langle \mathbf{a} + t\mathbf{b}, \mathbf{a} + t\mathbf{b} \rangle.$$

Using bilinearity,

$$\begin{aligned} f(t) &= \langle \mathbf{a}, \mathbf{a} \rangle + 2t\langle \mathbf{a}, \mathbf{b} \rangle + t^2\langle \mathbf{b}, \mathbf{b} \rangle \\ &= \|\mathbf{a}\|^2 + 2t\langle \mathbf{a}, \mathbf{b} \rangle + t^2\|\mathbf{b}\|^2. \end{aligned} \tag{1}$$

Since a squared norm is non-negative,

$$f(t) \geq 0 \quad \text{for all } t \in \mathbb{R}. \tag{2}$$

3. Discriminant argument. The right-hand side of (1) is a quadratic $qt^2 + pt + r$ with

$$q = \|\mathbf{b}\|^2, \quad p = 2\langle \mathbf{a}, \mathbf{b} \rangle, \quad r = \|\mathbf{a}\|^2.$$

If $\mathbf{b} \neq \mathbf{0}$ then $q > 0$. A quadratic with positive leading coefficient that never becomes negative must have a non-positive discriminant:

$$p^2 - 4qr \leq 0.$$

Substituting the expressions for p, q, r yields

$$(2\langle \mathbf{a}, \mathbf{b} \rangle)^2 - 4\|\mathbf{b}\|^2\|\mathbf{a}\|^2 \leq 0. \tag{3}$$

If $\mathbf{b} = \mathbf{0}$ then $f(t) = \|\mathbf{a}\|^2 \geq 0$ for every t , and (3) reduces to $0 \leq 0$. Hence (3) holds in all cases. Dividing (3) by 4 gives

$$\langle \mathbf{a}, \mathbf{b} \rangle^2 \leq \|\mathbf{a}\|^2 \|\mathbf{b}\|^2. \tag{4}$$

4. Returning to coordinates. Using the definitions from Step1, (4) becomes

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right),$$

which is the Cauchy–Schwarz inequality.

5. Equality case. Equality in (4) occurs precisely when the discriminant in (3) is zero.

If $\mathbf{b} \neq \mathbf{0}$, the discriminant being zero means that $f(t)$ has a double root; there exists a unique t_0 with $f(t_0) = 0$. Because a norm vanishes only for the zero vector, $\mathbf{a} + t_0 \mathbf{b} = \mathbf{0}$, i.e. $\mathbf{a} = -t_0 \mathbf{b}$. Setting $\lambda = -t_0$ gives $a_i = \lambda b_i$ for all i .

If $\mathbf{b} = \mathbf{0}$, both sides of the inequality are zero, so equality holds for every $\mathbf{a}$; the proportionality condition is satisfied trivially (take $\lambda = 0$).

Conversely, if $a_i = \lambda b_i$ for some $\lambda \in \mathbb{R}$, then $\mathbf{a} = \lambda \mathbf{b}$ and

$$\langle \mathbf{a}, \mathbf{b} \rangle = \lambda \|\mathbf{b}\|^2, \quad \|\mathbf{a}\|^2 = \lambda^2 \|\mathbf{b}\|^2,$$

so (4) becomes an equality. Hence equality holds *iff* the two families are proportional (including the degenerate case where one vector is zero). $\square$ $\square$